

Chapter 2 Assessment: MapReduce and the New Software Stack

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Student Information

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Academic Integrity Statement

By submitting this assessment, I affirm the following: I have completed this assessment independently, using only permitted course materials (lecture slides, textbook, and my own notes). I have not copied answers from another student, an online solution set, or an AI tool, and I have not given my answers to another student. I understand that submitting work that is not my own is a violation of academic integrity policy and may result in disciplinary action.

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Instructions

- Answer all questions in the space provided.
- Section D requires you to show your calculation steps, not just a final number.
- Where a question asks you to justify your answer, a one-line reason is sufficient unless stated otherwise.
- Estimated time: 45–60 minutes.

Section A: Multiple Choice (1 mark each)

A1. What property of "big data" makes it well-suited to parallel processing?

- ☐ (a) It is unstructured
- ☐ (b) It is extremely regular — the same operation applies to many similar records
- ☐ (c) It is always numerical
- ☐ (d) It fits in main memory

A2. In the new software stack, what is the role of the Distributed File System (DFS)?

- ☐ (a) To execute the Map and Reduce functions
- ☐ (b) To provide replication and redundancy against node/rack failure
- ☐ (c) To rank web pages by importance
- ☐ (d) To compress files before storage

A3. In a MapReduce job, which component is responsible for detecting a failed worker?

- ☐ (a) The DFS name node
- ☐ (b) Another worker process
- ☐ (c) The Master, via periodic pings
- ☐ (d) The user program directly

A4. Which scenario is a poor fit for MapReduce?

- ☐ (a) Computing a large matrix-vector multiplication
- ☐ (b) Running analytical aggregate queries over a huge, rarely-updated table
- ☐ (c) Managing a retail sales database with frequent small updates
- ☐ (d) Performing a large-scale relational join

A5. In the communication-cost model, why do we typically ignore output size and focus on input size?

- ☐ (a) Output is always zero
- ☐ (b) One task's output is another task's input, so it's already counted; output is also usually smaller than input
- ☐ (c) Output size cannot be measured
- ☐ (d) Only Map tasks produce meaningful output

Section B: True / False (1 mark each — briefly justify each answer)

B1. A single file in a distributed file system is typically stored as one large, unbroken block.

☐ True ☐ False

Justification:

B2. MapReduce's Reduce function receives all the values associated with a single key at once.

☐ True ☐ False

Justification:

B3. Designing a good MapReduce algorithm is essentially the same skill as designing a good algorithm for a supercomputer.

☐ True ☐ False

Justification:

B4. A multiway join can, in some cases, have lower total communication cost than performing the equivalent sequence of two-way joins.

☐ True ☐ False

Justification:

B5. If a compute node fails mid-task, MapReduce recovers the partial results already computed by that node rather than restarting the task.

☐ True ☐ False

Justification:

Section C: Short Answer (2–3 marks each)

C1. Explain, in your own words, the difference between the loss of a single node and the loss of an entire rack in a compute cluster, and why file systems must design around both.

C2. Describe what happens, step by step, from the moment a Map task emits a (key, value) pair to the moment the corresponding Reduce task begins processing it.

C3. Give one example (not from the slides) of a real-world computation that would be a good fit for MapReduce, and one that would be a poor fit. Justify each in one sentence.

C4. In the relational-algebra operations covered in this chapter, explain the difference between selection and projection, using your own example (not the Friends table).

Section D: Problem Solving (show your work)

D1. Word Count Trace

Given two input chunks — Chunk 1: "data mining is fun" | Chunk 2: "data science is data driven"

(a) List all the (key, value) pairs emitted by the Map step.

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(b) After grouping by key, list the values each Reduce task receives.

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(c) Give the final Reduce output for every key.

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D2. Matrix-Vector Multiply

Given $M = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{bmatrix}$ and $v = [1, 2, 1]$

(a) Write out the (key, value) pairs the Map step would emit for row $i=2$.

(b) Compute the full product Mv by hand, showing your work for each row.

D3. Communication Cost of a Join

Relation $R(A,B)$ has 2,000 tuples and relation $S(B,C)$ has 800 tuples. Each tuple is approximately 120 bytes.

(a) Compute the communication cost of the join $R \text{ JOIN } S$, in number of tuples. (b) Compute the same cost in bytes.

D4. Multiway Join vs. Chained Two-Way Joins

Given relations $R(A,B)$, $S(B,C)$, $T(C,D)$ with sizes $r=400$, $s=250$, $t=350$ tuples.

(a) Compute the communication cost of a single one-pass multiway join $R \text{ JOIN } S \text{ JOIN } T$.

(b) Suppose computing $R \text{ JOIN } S$ first produces an intermediate result of 300 tuples. Compute the total cost of doing this as two sequential two-way joins.

(c) Which approach is cheaper here, and by how much?

D5. DFS Chunking

A dataset file is 500MB, and the DFS uses a chunk size of 64MB.

(a) How many chunks will this file be split into? Show the rounding logic. (b) If each chunk is replicated 3 times across different racks, how many total chunk-copies exist for this file?

Section E: Applied / Design Thinking (open-ended, 3–5 marks each)

E1. You are asked to compute, for a social network with 500 million users, the average number of friends per user. Sketch how you would structure this as a MapReduce job: what does the Map function emit, and what does the Reduce function compute? Which relational-algebra operation does this correspond to?

E2. Suppose the vector v in a matrix-vector multiply is too large to fit in main memory. Explain, in your own words, the striping strategy used to handle this, and why it solves the memory problem.

E3. A colleague argues: "Since MapReduce tolerates failures automatically, we should use it for every data processing task in our company, including our live order-management database." Do you agree or disagree? Justify your answer using at least two concepts from this chapter.

End of Assessment — please review your Academic Integrity acknowledgment on page 1 before submitting.